OpenVisualJev: Design and Analysis of a Multimodal Decision Model

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Abstract

Multimodal decision models support low-latency decisions by mapping visual context and runtime-defined choices directly to probabilities. We present OpenVisualJev, a vision–language model with tree-masked candidate encoding, permutation-equivariant set interaction, and a dynamic pointer head. Shared image and question prefixes provide context for isolated candidate branches; joint comparison then produces set-dependent scores that remain consistent under candidate reordering. Calibration-aware training and held-out temperature scaling refine these probabilities. Experiments on visual relations, fine-grained recognition, and natural VQA demonstrate order stability, candidate-dependent comparison, and competitive standard-task performance, with larger gains on difficult candidate sets. Ablations identify the contributions of candidate interaction, variable-count training, and calibration. Code is available at `https://github.com/jin-s13/OpenVisualJev`.

1 Introduction

Vision–language models support general-purpose visual understanding, while decision pipelines require low-latency judgments over runtime-defined alternatives. The System One paradigm addresses this requirement through direct probabilistic prediction, extending naturally to image-conditioned decisions (Yu & Yao, 2026; Zhou et al., 2026).

Candidate scoring must accommodate both order symmetry and set dependence: reordering should preserve candidate identity, while changing the alternatives may change relative preferences. Shared encoding and joint comparison combine these properties through set representations (Zaheer et al., 2017; Lee et al., 2019).

We present OpenVisualJev[1], which separates candidate grounding from joint comparison. Tree-masked attention shares image and question prefixes across isolated candidate branches in one forward pass. A position-free set module compares their representations, and a shared pointer produces candidate probabilities. Calibration-aware training and held-out temperature scaling complete the model (Figure 1).

Controlled order and relation tests examine the architecture's structural properties; unified recognition and natural VQA assess task performance. Ablations isolate candidate interaction, training with variable candidate counts, and calibration.

Our contributions are threefold:

- A multimodal decision model combining pretrained visual representations, dynamic candidate scoring, and probability calibration.
- A permutation-equivariant scoring architecture with joint candidate comparison, together with an analysis of its structural properties.
- Controlled structural tests, recognition and VQA evaluations, and ablations of interaction, candidate-count training, and calibration.

[1] An independently developed research implementation of a Jev-style multimodal decision model.

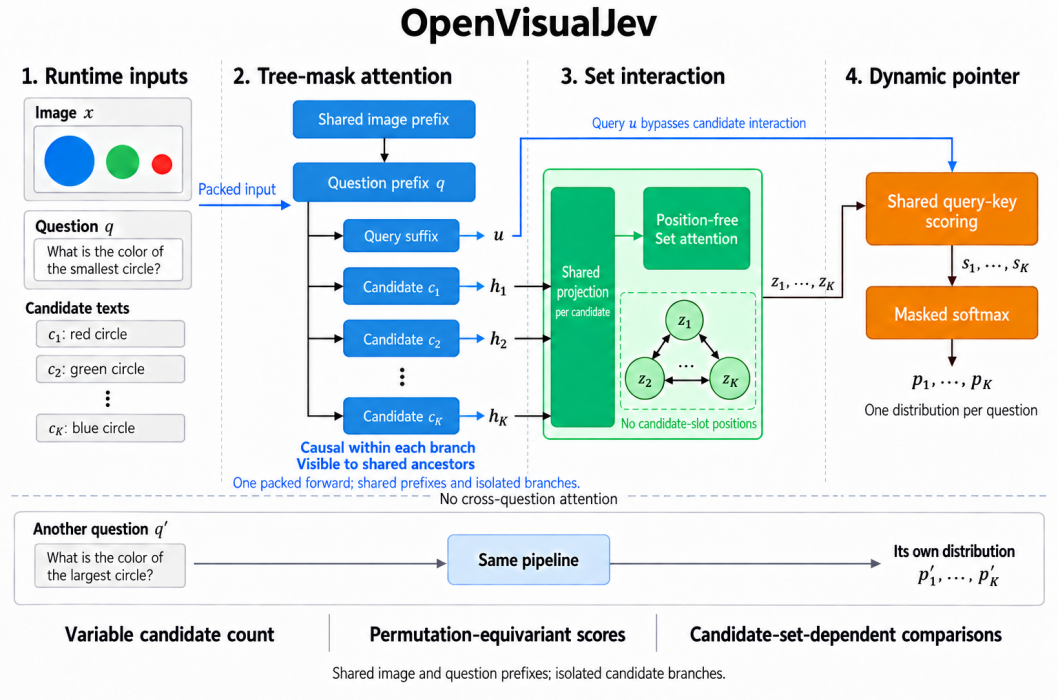


Figure 1: OpenVisualJev. Tree-masked attention shares image and question prefixes across isolated query and candidate suffixes. Tokens attend to causal predecessors and ancestors. Position-free Set attention compares candidate representations; the query bypasses this module and scores its outputs through a shared pointer. Interaction and normalization are question-local. Training-time confidence regularization and held-out temperature scaling calibrate the output probabilities.

2 Decision interface and model structure

Given an image x, a question q, and runtime candidate texts $C = (c_i)_{i=1}^{K}$, OpenVisualJev returns a normalized distribution over mutually exclusive candidate slots. External identifiers associate outputs with inputs; prompts contain image, question, and candidate content.

2.1 Tree-masked multimodal encoding

The input tree contains a shared image prefix, one prefix per question, and separate query and candidate suffixes. Each prefix appears once. Tokens attend to causal predecessors within their segment and to ancestors; sibling candidates and different questions remain isolated.

Visibility and positions. Let $\mathcal{A}(t)$ contain tokens in the ancestor segments of token t, and let $\mathcal{C}(t)$ contain tokens up to and including t within its own segment. The binary visibility mask is $M_{ts} = \mathrm{I}[s \in \mathcal{A}(t) \cup \mathcal{C}(t)]$. The shared root is causal; a question prefix reads the root and its own history; each leaf additionally reads its question prefix. Masked entries are excluded from attention normalization. Packed offsets only locate tokens in memory: multimodal position coordinates follow the original root-to-leaf sequence. Consequently, placing one candidate before another in the packed sequence creates no additional attention edge between them.

Branch representations. Let E_θ return the final valid-token representation along a root-to-leaf path. The query yields $u = E_\theta(x, q)$, and candidate i yields $h_i = E_\theta(x, q \parallel \texttt{Candidate:} \parallel c_i)$. One packed forward computes all leaves, injecting visual features at the

shared root. Original branch-local multimodal positions preserve each candidate's coordinates and visible context under reordering.

For multiple questions about one image, each question receives its own internal prefix and leaf group beneath the shared root. Final-token representations are gathered by leaf identity and passed to separate question-local scoring operations. Adding another question therefore adds a disjoint subtree while retaining the existing questions' visible tokens. Candidate count determines the number of leaves and the size of the subsequent set operation; the encoder and pointer parameters are shared across counts.

2.2 Set interaction and dynamic pointer

Position-free candidate interaction. Stacking candidate representations as rows of H, we apply a shared projection and position-free self-attention module S_ϕ. The query bypasses this module. Scores and probabilities are

$$Z = S_\phi(H), \qquad s_i = \frac{\langle W_q \,\mathrm{LN}(u), W_k \,\mathrm{LN}(Z_i)\rangle}{\sqrt{d}}, \qquad p_i = \frac{\exp(s_i)}{\sum_{j\in\mathcal{V}} \exp(s_j)}, \tag{1}$$

where d is the query–key dimension and $\mathcal{V}$ indexes valid slots. Padding is excluded from attention and probability normalization.

2.3 Structural properties

Permutation-equivariant scoring. Shared row-wise operations commute with a candidate permutation P. Consistently permuting query, key, value, and mask gives $\mathrm{Attn}(PH) = P\,\mathrm{Attn}(H)$; residuals and row-wise normalization preserve this relation. Thus $s(PH) = Ps(H)$ and $p(PH) = Pp(H)$ in exact arithmetic, provided branch preprocessing and positions are unchanged and candidate indices are omitted.

With a unique maximum, the selected candidate identity is preserved under candidate reordering.

The attention relation follows from permuting both axes of the attention matrix: if $A = \mathrm{softmax}(QK^\top/\sqrt{d})$, then the permuted matrix is $PAP^\top$. Multiplication by the permuted values PV gives PAV. This identity extends through each candidate block, while the shared query remains unchanged. The pointer therefore permutes output slots together with their candidate identities.

Question isolation and candidate-set dependence. Tree masking isolates questions, and Set attention and normalization operate within each question. Within a question, logits depend on all candidates, so $p_i/p_j = \exp(s_i - s_j)$ can change when the candidate set changes.

2.4 Training and probability calibration

Calibration-aware training. We augment cross-entropy with dataset-wise confidence–accuracy alignment, following DCA (Liang et al., 2020). For dataset groups $B(d)$ in an effective batch, define $a_n = \arg\max_i p_{n,i}$, confidence $c_n = \max_i p_{n,i}$, and correctness $r_n = \mathrm{I}[a_n = y_n]$. Over represented datasets $\mathcal{D}$, we optimize

$$\begin{aligned} \mathcal{L}_{\text{train}} &= \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{conf}}, \\ \mathcal{L}_{\text{conf}} &= \frac{1}{|\mathcal{D}|} \sum_{d\in\mathcal{D}} \left[\frac{1}{|B(d)|} \sum_{n\in B(d)} \left(c_n - \mathrm{stopgrad}(r_n)\right) \right]^2 . \end{aligned} \tag{2}$$

Balanced batches weight datasets equally. Gradients flow through confidence while correctness is detached. The loss ablation sets $\lambda = 0$ under matched sampling and optimization.

The two loss terms operate at different levels. Cross-entropy supervises each target candidate, whereas the confidence term aligns mean confidence and accuracy within each dataset.

Equal group weighting prevents datasets with more examples in an effective batch from dominating that alignment. Post-training temperature fitting then adjusts the frozen model's probability scale on a separate data split.

Reserved probability calibration. With model weights frozen, we fit $T > 0$ on reserved calibration data by minimizing candidate-event NLL. Predictions use $p_i^{(T)} = \exp(s_i/T)/\sum_{j\in\mathcal{V}} \exp(s_j/T)$. Scalar scaling preserves ranking, candidate identity, and permutation equivariance.

3 Experimental setup

All methods use Qwen3-VL-4B-Instruct (Bai et al., 2025), frozen vision parameters, and LoRA on language attention projections (Hu et al., 2021). Set Pointer and the complete-candidate LM are trained with AdamW (Loshchilov & Hutter, 2017), cross-entropy, and the dataset-wise calibration loss. Independent Pointer additionally tests candidate-independent scoring in the relation task. Section 4.1 defines the interfaces.

The reported accuracy and probability tables were evaluated with ordinary batched branches using the same branch visibility and positional coordinates described in the method.

Architecture and scoring. The backbone hidden width is 2560. Candidate summaries are projected to width 256; Set uses two pre-normalized position-free Transformer blocks, four attention heads, feed-forward expansion four, GELU, and zero dropout. Independent removes cross-candidate blocks. The separate question query bypasses candidate attention. The readout operates in single precision over bfloat16 backbone features. The LM scores the next-token logits of single-token candidate symbols, directly normalized over the supplied alternatives. Pointer branches omit candidate indices. Pointer uses separate candidate contexts, while LM uses one complete-candidate context.

Unified recognition. One checkpoint per method covers Pets (Parkhi et al., 2012), CUB, Cars (Krause et al., 2013), Dogs, Flowers (Nilsback & Zisserman, 2008), and Aircraft (Maji et al., 2013). Training, development, and calibration contain 31,088, 3,637, and 3,635 images. Testing uses 35,566 FOCI-style four-choice records (Geigle et al., 2024) released with Finedefics (He et al., 2025), preserving its repeated Dogs row and shuffling candidates identically across methods.

Data and candidate construction. Development and calibration images are reserved from official training-side data, using the official validation images for Flowers and Aircraft. Exact and reviewed near-duplicate image groups are separated from the test inventory, with cross-dataset exclusions before mixed training. The six datasets contain 37, 200, 196, 120, 102, and 100 classes, respectively. Each classification record supplies a smaller candidate set containing the gold class. Standard testing preserves the released Finedefics candidates. For difficult-candidate transfer, incorrect classes are selected by a fixed OpenCLIP ViT-L-14 image–text ranking, so each model receives the same distractors at each candidate count.

Main model and component ablation. The complete models use balanced hard-four-choice training: 96 questions per effective batch, 16 per dataset, seed 20261021, and the epoch-three checkpoint after 2,016 updates. Both receive identical candidate exposures, calibration-loss weight five, and separately fitted temperatures minimizing equal-dataset calibration NLL. Table 7 ablates the training penalty and temperature under this recipe.

The learning rate is 0.0001 with 61 warmup updates, cosine decay, AdamW weight decay 0.01, gradient clipping at norm one, and language-attention LoRA rank/scaling 16/32. Training comprises 193,536 question exposures. The calibration coefficient of five is the evaluated development setting; the balanced recipe was developed after inspecting earlier benchmark results. The CE control sets this coefficient to zero, retaining the same data, updates, and final-epoch selection. Each model fits a positive scalar temperature on the reserved calibration split before standard testing.

Natural VQA. Both interfaces are separately adapted to A-OKVQA (Schwenk et al., 2022) across three seeds and evaluated on 2,576 internal holdout questions. Training, selection, and calibration roles are disjoint.

VQA splits and training. A-OKVQA training uses 12769 questions over 12404 images; selection uses 859 questions over 827 images, and calibration uses 852 questions over 827 images. The internal holdout has 2576 questions over 2482 images, grouped into 2481 resampling components after a reviewed near-duplicate merge. It is split from the official training inventory. Seeds are 20260924, 20260927, and 20260930. Models train for three epochs with learning rate 0.0002 and accumulation over eight questions; LoRA, optimizer, warmup, and clipping settings otherwise follow the shared recipe above. Checkpoint selection maximizes answer-text correctness, then minimizes NLL, then prefers the earlier epoch. Blank and duplicate choices are retained; probability mass for duplicate literal answer strings is summed when computing answer-event losses. Images are uncropped.

Metrics. We report accuracy, NLL, sum-Brier, and ECE with held-out temperature calibration (Guo et al., 2017). Recognition weights datasets equally. Paired intervals resample image clusters for recognition and scenes for relations (Appendix A). Bold marks best point estimates, including displayed ties.

Selection and uncertainty. Complete-method comparisons use the fixed third-epoch checkpoints. The separate CE candidate-count ablation selects checkpoints by equal-dataset development accuracy, preferring the earlier epoch on ties. Calibration and test examples have separate roles, and fitted temperatures remain frozen during testing. Recognition intervals use 10,000 paired bootstrap replicates within datasets, retaining repeated image rows in the same cluster and averaging datasets equally. Relation intervals resample complete scenes to preserve paired questions. VQA intervals resample images and the three training seeds and are exploratory. Intervals condition on the selected weights and fitted temperatures; they describe evaluation-sample uncertainty rather than variation over retraining or calibration refitting. All reported intervals are marginal, without multiplicity correction.

4 Results and analysis

4.1 Order stability

We compare three interfaces to test candidate-order stability (Table 1).

Independent Pointer scores isolated candidate representations with a shared 256-dimensional query–key pointer. **Set Pointer** adds two position-free Transformer blocks (four heads, fourfold feed-forward expansion, GELU, zero dropout); the query bypasses interaction. **Complete-candidate LM** jointly encodes labeled alternatives and normalizes next-token logits over their single-token labels. All directly output candidate probabilities.

Interface	Candidate interaction	Equivariant scores
Independent Pointer	No	Yes
Set Pointer	Yes	Yes
Complete-candidate LM	Yes	Not guaranteed

Table 1: Structural properties under the preprocessing and masking conditions in the method. Equivariance is a score property; ties and floating-point effects require separate decision-level checks.

On 755 questions, we evaluate all 24 candidate permutations and align predictions by candidate identity. LM changes its answer on 108 questions (14.30%); Set changes on none. Figure 2 summarizes order stability and difficult-candidate transfer for the complete-method checkpoints.

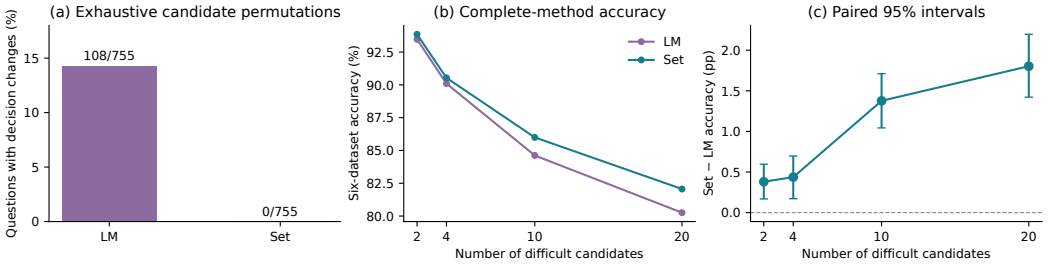


Figure 2: Complete-method structural analysis. (a) Fraction of questions whose predicted candidate identity changes across all twenty-four orders. (b) Six-dataset mean accuracy on difficult candidate sets. (c) Set-minus-LM accuracy with marginal paired image-cluster bootstrap intervals. All panels use the same complete-method checkpoints as Table 3; the training candidate count is four.

4.2 Candidate interaction enables preference reversals

Paired three-choice questions replace an extreme object while retaining two shared objects, reversing their correct median ranking. We evaluate fixed checkpoints on 512 fresh scenes (2,048 questions), averaging three seeds; a maximum-object control tests ordinary ranking. Training updates language LoRA and the trainable pointer head using choice-label supervision. The original training inventory has 256 scenes; the final test uses fresh scenes to evaluate the fixed training endpoint. Seeds are 20261011, 20261012, and 20261013. The common fixed endpoint is 80 epochs, using AdamW learning rate 0.0003, accumulation over 64 questions, weight decay 0.01, and clipping at norm one. This longer-budget test followed earlier development diagnostics; it evaluates the final achievable capability under this recipe. All final seeds are retained.

	Median relation			Maximum control	
Model	Acc. (%)	Pairs (%)	NLL	Acc. (%)	NLL
Independent	50.0000	0.0000	0.70072	99.7721	0.01175
Set	99.0234	98.3073	0.25971	99.5768	0.03669
Complete LM	**99.7721**	**99.5443**	**0.01878**	**99.8372**	**0.01062**

Table 2: Fresh-scene test of fixed 80-epoch checkpoints, equally averaging the three retained training seeds. Pairs require both questions correct; all NLL values are uncalibrated. Set and complete-candidate LM both learn the median relation, which requires candidate interaction. The training budget is matched by optimizer updates.

Set exceeds Independent by 98.3073 points in paired median accuracy (scene interval $[97.4609, 99.0234]$). Independent succeeds on the maximum control, supporting the role of candidate interaction.

For Independent, a candidate has utility $f(x, q, c_i)$, so the log odds between two shared candidates are $f(x, q, c_i) - f(x, q, c_j)$. Replacing a third candidate changes the softmax denominator but preserves these log odds and their ordering. The paired median test requires the ordering to reverse. Set can satisfy this requirement because its logits condition on the entire candidate collection. The maximum control verifies that the task also contains visual comparisons that independent utilities can learn.

4.3 Standard recognition and candidate-count transfer

After matched training and held-out calibration, Set reaches 90.35% mean accuracy versus 90.09% for LM (Table 3; difference +0.2690 points, paired interval $[+0.0086, +0.5317]$).

Dataset	LM (%)	Set (%)	Difference (pp)
Pets	**95.88**	95.42	-0.46
CUB	84.52	**85.57**	+1.05
Cars	93.74	**93.83**	+0.09
Dogs	85.65	**85.80**	+0.15
Flowers	**99.11**	98.99	-0.11
Aircraft	81.61	**82.51**	+0.90
Equal-dataset mean	90.09	**90.35**	+0.27

Table 3: Six-dataset accuracy on identical author-defined four-choice records. Each method uses one balanced CE-plus-calibration checkpoint and a held-out temperature. One training seed; differences are Set minus LM before rounding.

Transfer to larger difficult candidate sets. With fixed checkpoints and image–text-ranked distractors, the Set–LM gap grows from 0.44 points at four choices to 1.80 at twenty (Table 4; paired intervals $[0.17, 0.70]$ and $[1.42, 2.20]$).

Candidates	LM (%)	Set (%)	Difference (pp)
2	93.47	**93.85**	+0.38
4	90.10	**90.54**	+0.44
10	84.62	**86.00**	+1.38
20	80.26	**82.06**	+1.80

Table 4: Difficult-candidate transfer of the four-choice-trained models in Table 3. Test distractors use a fixed image–text ranking. Values are six-dataset mean accuracies; differences are Set minus LM.

This test changes the decision set while holding the image, category inventory, and checkpoint fixed. It therefore evaluates sensitivity to the number and difficulty of alternatives. The separate variable-count experiment below asks whether exposing the model to such variation during training improves this behavior. Keeping these comparisons separate distinguishes inference-time candidate expansion from changes to the training distribution.

4.4 Training with variable candidate counts

Fixed four-choice and variable-K training ($K \in \{2, 4, 10\}$, uniformly sampled) share image exposure, updates, development selection, and test sets. Candidate-token computation varies; twenty-choice tests exceed the training range. Variable-K training improves twenty-

Test	LM fixed	Set fixed	LM variable	Set variable
Standard	89.08	89.79	89.08	**89.82**
Hard 2	92.87	**93.36**	92.73	93.26
Hard 4	89.25	**89.93**	89.13	89.91
Hard 10	84.94	86.27	86.02	**87.24**
Hard 20	82.01	84.47	84.20	**85.82**

Table 5: Equal-dataset accuracy (%) for fixed four-candidate training and variable-candidate training. All four models use the same single training seed and test records. Difficult-distractor tests and the author-defined standard test are separate constructions.

choice accuracy by 2.19 points for LM and 1.35 for Set. Set retains a 1.62-point lead (paired interval $[1.31, 1.93]$) and perfect tested order stability; standard accuracy changes little.

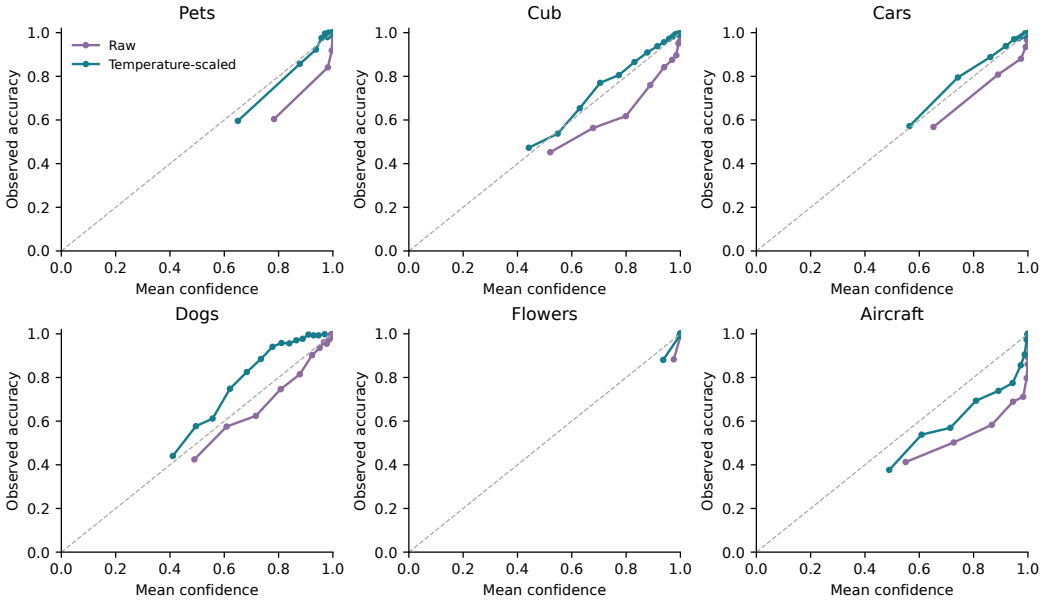


Figure 3: Calibration of the complete Set model. Points show mean confidence and accuracy in fifteen equal-count bins; the diagonal denotes agreement. One temperature fitted on reserved data adjusts all six datasets while preserving predictions.

4.5 Training and post-training calibration

Temperature scaling improves both models' NLL and Brier without changing accuracy (Table 6). Scaled NLL favors LM; Set's lower Brier point estimate has a paired difference interval containing zero. Figure 3 shows the shared temperature's effect across datasets.

	Raw		Temperature-scaled	
Metric ↓	LM	Set	LM	Set
NLL	0.3008	0.3478	**0.2778**	0.2831
Sum-Brier	0.1473	0.1480	0.1438	**0.1428**
ECE	0.0337	0.0448	**0.0224**	0.0355

Table 6: Equal-dataset mean probability metrics for the balanced CE-plus-calibration checkpoints in Table 3. ECE uses fifteen equal-count bins and is descriptive. Lower is better; bold marks the lowest value in each row.

Training-time calibration. The training penalty improves raw NLL, Brier, and ECE. Combined with temperature scaling, it reduces NLL by 0.01539 (paired interval $[-0.01953, -0.01142]$) relative to scaled CE, with a -0.1273-point accuracy change (Table 7).

4.6 Natural VQA performance

Set retains comparable VQA accuracy: the three-seed mean decreases by 0.53 points relative to native LM (exploratory interval $[-1.7041, 0.6406]$). Scaled probability losses favor LM; Table 8 reports the results.

Training	Probability	Accuracy (%)	NLL ↓	Brier ↓	ECE ↓
CE	Raw	**90.48**	0.4003	0.1519	0.0521
CE + calibration	Raw	90.35	0.3478	0.1480	0.0448
CE	Temperature	**90.48**	0.2985	0.1457	0.0427
CE + calibration	Temperature	90.35	**0.2831**	**0.1428**	**0.0355**

Table 7: Training-loss comparison for Set under balanced six-dataset training with one shared seed. Metrics are equal-dataset means on the standard test; Brier is summed over candidates and ECE uses fifteen equal-count bins. Each model receives its own held-out temperature fitted with an equal-dataset NLL objective. Bold marks the best value in each metric column, including ties.

Interface	Accuracy (%)	Scaled NLL	Scaled Brier
Native LM	**87.2800**	**0.3703**	**0.1845**
Set	86.7495	0.3819	0.1932

Table 8: Natural VQA capability and calibrated probability quality, using the completed supervised holdout experiment. Means use all registered seeds and frozen, independently fitted temperatures. The comparison follows the registered native-LM secondary analysis. Raw probability differences and protocol details appear in Appendix A.

5 Related work

Multimodal decision models. Visual Jev and PixelJev use option-token readouts (Yu & Yao, 2026; Zhou et al., 2026); Koa-action learns atomic labels under randomized mappings (Dai et al., 2026). OmniJev combines learned scores, probabilistic supervision, and calibration (Xu et al., 2026), while Chinese-Jev uses a text encoder and heterogeneous supervision (Wang et al., 2026). Visual Jev shares image prefixes, and FastJEV studies state reuse and layer pruning (Yu & Yao, 2026; Ma et al., 2026). OpenVisualJev combines isolated candidate grounding with permutation-equivariant joint comparison.

Set-dependent choice. Deep Sets and Set Transformer provide invariant and equivariant representations with learned interactions (Zaheer et al., 2017; Lee et al., 2019). FETA and FATE model utilities that depend on the surrounding alternatives (Pfannschmidt et al., 2019). We apply these principles to multimodal candidates and test both preference reversals and order stability.

Probability quality and decision evaluation. Decision benchmarks examine calibration, selective prediction, ordering, and candidate counts (Deußer et al., 2026; Rafe & Das, 2026); routing, security, and mobile agents use such scores downstream (Li et al., 2026; Liu, 2026; Zhang, 2026). Visual MCQA debiasing and selective VQA study related reliability issues (Atabuzzaman et al., 2025; Eisenschlos et al., 2024). Our probability pipeline combines confidence–accuracy regularization (Liang et al., 2020; Hebbalaguppe et al., 2022) with temperature scaling (Guo et al., 2017).

Fine-grained visual recognition. FOCI casts fine-grained recognition as multiple-choice prediction (Geigle et al., 2024); Finedefics combines attribute alignment with classification instruction tuning (He et al., 2025). We reuse its released candidate records to compare unified Qwen3-VL scoring architectures across six datasets.

6 Conclusion

OpenVisualJev combines shared multimodal encoding, permutation-equivariant comparison, and calibrated pointer scoring. Structural tests verify order stability and set-dependent preferences; recognition and VQA establish task capability. Candidate-count training and calibration further improve robustness and probability quality.

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AI usage disclosure

Under the author's direction, generative AI and the `ai-research-writing-skill` workflow assisted literature review, research design, implementation, analysis, figures, and writing. Executable code generated synthetic data; model predictions and evaluation artifacts supplied the reported results. The author reviewed the work and retains responsibility for its claims, code, data, and figures.

A Essential reproducibility details

Recognition data and optimization. Pets, CUB, Cars, and Dogs reserve development/calibration images within classes from the official training-side inventory. Flowers and Aircraft use official validation images for those roles. Exact and reviewed near-duplicate groups are separated from test-side images; additional cross-dataset exclusions precede mixed training. The duplicate audit covers the task-specific training and evaluation inventories. The six datasets contain 37, 200, 196, 120, 102, and 100 classes, respectively; the choice count is independent of class count.

FOCI/Finedefics test candidate sets are preserved, including the original repeated Dogs row, and shuffled consistently across methods. The unified protocol follows the released Dogs test inventory. Training and dynamic tests mine difficult incorrect labels with the recorded OpenCLIP ViT-L-14 image–text ranking; random distractors are distinct sampled incorrect classes. The correct label is included. The standard author-defined four-choice test and the newly constructed difficult four-choice test are separate conditions.

Unified training uses learning rate 0.0001, accumulation over eight questions, three epochs, cosine decay, warmup fraction 0.03, AdamW weight decay 0.01, global clipping at norm one, and language-attention LoRA rank/scaling 16/32. The seed is 20261021. Both methods use candidate-conditioned cross-entropy, the same exposure and selection rule, and development-based checkpoint selection. Variable-count training changes only the registered candidate-count sampling and associated candidate construction; candidate-token computation follows the sampled candidate counts. Images use processor area bounds 65536 and 262144 pixels; Aircraft crops the bottom twenty pixels. Other recognition images are uncropped.

Relation construction and endpoint. Images contain colored objects with distinct sizes. Paired three-choice questions share two middle objects but replace the third candidate with a smaller or larger extreme, reversing the median answer. The largest-object task serves as an independent-ranking control. Whole-scene bootstrap intervals condition on those trained seeds.

VQA comparison scope. The main text presents the registered native-LM secondary comparison. The original primary contrast used a canonically sorted LM input: Set-minus-canonical-LM accuracy was -0.5694 percentage points, with conditional interval $[-1.2861, 0.1289]$. The original primary and secondary analysis roles are retained. For

native LM, raw Set-minus-LM NLL and Brier differences are +0.1933 and +0.0243; their exploratory crossed intervals are $[0.0779, 0.3260]$ and $[0.0078, 0.0407]$. Scaled differences are +0.0116 for NLL and +0.0087 for Brier, with crossed intervals $[-0.0054, 0.0284]$ and $[-0.0025, 0.0204]$. The registered analysis estimates accuracy and probability-loss differences with intervals.

Calibration and statistical conventions. Recognition minimizes reserved-set pooled candidate NLL over $\log T \in [-5, 5]$ using bounded scalar optimization and explicit boundary checks. VQA uses a fixed grid of 2001 log-temperatures in the same range, preferring the value closest to zero on objective ties. Temperatures are fixed before test scoring. Sum-Brier sums squared probability errors across candidate or literal-answer events. Recognition ECE uses fifteen equal-count bins; VQA uses fifteen equal-width bins. Exact score ties are resolved by selecting the first slot. Paired bootstrap samples keep repeated images and related questions together. The recognition and relation intervals condition on the trained seeds; the crossed VQA analysis is exploratory with few training realizations. All intervals are conditional on the fitted temperatures and the evaluation distribution.

Reproduction scope. The recorded implementation uses PyTorch 2.8.0, Transformers 4.57.3, PEFT 0.17.1, and NVIDIA A800 devices. Data manifests, model checkpoints, raw predictions, and analysis scripts provide the evidence trail. The project archive preserves earlier specialist adapters, intermediate training endpoints, optimization probes, and engineering validation records. Public external reproduction remains a planned release milestone.

A.1 Equivalent answer events

For the registered VQA protocol, blank and duplicate candidate strings are preserved. Prediction selects the maximum-logit slot, with first-slot tie breaking, before comparison by literal answer text. For probability scoring, the mass of identical strings is summed. Given a gold answer event G, the candidate-conditioned loss is

$$\mathcal{L} = -\log \sum_{i \in G} p_i = \mathrm{LSE}_{j \in \mathcal{V}}(s_j) - \mathrm{LSE}_{i \in G}(s_i). \tag{3}$$

Labels supply supervision to the loss and evaluator; forward prompts contain the image, question, and candidate text. This convention preserves the official answer-text task while keeping slot and answer-event probabilities distinct.

A.2 Supplementary evaluation details

Candidate-count study. The variable-count training ablation uses a separate unified CE recipe: three epochs with matched image exposure, alternating difficult and random distractors, and development selection by equal-dataset accuracy with earlier-epoch tie-breaking. Fixed-count training uses four candidates; variable-count training samples uniformly from $K \in \{2, 4, 10\}$. The reported count-transfer tests use difficult distractors at two, four, ten, and twenty candidates alongside the standard records. The four checkpoints share the training seed and test manifests. This study matches image exposure and updates; candidate-token computation varies with candidate count. Twenty-candidate testing measures transfer beyond the training count range.

Order and relation tests. The order test evaluates all 24 permutations of four candidates on 755 class-stratified questions, aligning candidate identities before comparison. Both this test and the difficult-candidate count-transfer evaluation use the complete-method checkpoints from Table 3, with unchanged weights and inference precision. Relation tests use synthetic scenes with paired candidate sets that reverse the required preference between shared objects. Independent, Set, and complete-candidate LM are evaluated with three training seeds at the same fixed final training endpoint on fresh scenes. The paired metric requires both questions correct; a largest-object control tests ordinary independent ranking.

Probability quality and uncertainty. Each model receives its own scalar temperature fitted only on reserved calibration data (Guo et al., 2017). We report accuracy, NLL, sum-Brier, and ECE, distinguishing raw and scaled probabilities. Recognition ECE uses fifteen equal-count bins; the VQA protocol uses equal-width bins over literal answer events. Recognition intervals use 10,000 paired image-cluster bootstrap replicates within datasets and equal dataset weighting; repeated image rows stay together. They condition on the selected weights and fitted temperatures, excluding training-seed and calibration-fit uncertainty. Relation intervals resample whole scenes; the VQA interval additionally resamples the few available seeds and is exploratory. Intervals are marginal, without multiplicity correction. Further protocol details appear in Appendix A.